

In-silo federated learning vs. centralized learning for segmenting acute and chronic ischemic brain lesions[☆]

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ABSTRACT

Objectives: To investigate the efficacy of federated learning (FL) compared to industry-level centralized learning (CL) for segmenting acute infarct and white matter hyperintensity.

Materials and methods: This retrospective study included 13,546 diffusion-weighted images (DWI) from 10 hospitals and 8421 fluid-attenuated inversion recovery (FLAIR) images from 9 hospitals for acute (Task I) and chronic (Task II) lesion segmentation. We trained with datasets originated from 9 and 3 institutions for Task I and Task II, respectively, and externally tested them in datasets originated from 1 and 6 institutions each. For FL, the central server aggregated training results every four rounds with FedYogi (Task I) and FedAvg (Task II). A batch clipping strategy was tested for the FL models. Performances were evaluated with the Dice similarity coefficient (DSC).

Results: The mean ages (SD) for the training datasets were 68.1 (12.8) for Task I and 67.4 (13.0) for Task II. The frequency of male participants was 51.5 % and 60.4 %, respectively. In Task I, the FL model employing batch clipping trained for 360 epochs achieved a DSC of 0.754 ± 0.183 , surpassing an equivalently trained CL model (DSC 0.691 ± 0.229 ; $p < 0.001$) and comparable to the best-performing CL model at 940 epochs (DSC 0.755 ± 0.207 ; $p = 0.701$). In Task II, no significant differences were observed amongst FL model with clipping, without clipping, and CL model after 48 epochs (DSCs of 0.761 ± 0.299 , 0.751 ± 0.304 , 0.744 ± 0.304). Few-shot FL showed significantly lower performance. Task II reduced training times with batch clipping (3.5–1.75 h).

Conclusions: Comparisons between CL and FL in identical settings suggest the feasibility of FL for medical image segmentation.

1. Introduction

Numerous studies have investigated machine learning (ML) for segmenting medical images, demonstrating promising performance [1–3]. Typically, these studies depend on preprocessed and aggregated open-source medical images. Injecting these algorithms into clinical practice is limited, however, partly due to suboptimal performance and domain shift [4]. To address the representation gaps in open datasets, it is crucial to access diverse datasets from a broader range of imaging

vendors and institutions. Unfortunately, legal constraints pose significant challenges for institutions and hospitals in sharing images with third-party servers for model training.

Unlike ML methodologies that depend on centralized data for training, federated learning (FL) [5,6] aggregates outcomes from individually trained models across multiple data silos. Upon receiving the model, clients train their local models with their private data. Upon completion, clients send their local model parameters back to the central server. The server then aggregates these submissions via an aggregation method to construct the next generation of the central model, which is

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Abbreviations

ML	Machine Learning
FL	Federated Learning
CL	Centralized Learning
DWI	Diffusion Weighted Imaging
FLAIR	Fluid Attenuated Inversion Recovery
WMH	White Matter Hyperintensity
ANOVA	Analysis of Variance
DSC	Dice Similarity Coefficient
SD	Standard Deviation

sent it back to the clients for performance evaluation and declaration of a new round. By design, FL allows hospitals, acting as clients, to retain full privacy of their data while facilitating parallel, collaborative training across all partitioned datasets. In effect, FL achieves the goals of ML without risking potential leakage of medical data from the centralized third-party server. With such benefits, FL emerged as a compelling alternative to centralized learning (CL) in applicative fields including of medical imaging.

Despite its advantages, FL is often underestimated due to concerns about potential underperformance compared to CL [7]. One of the most significant limitations of FL is the potential of non-independent and identically distributed (non-i.i.d.) data [8]. In practice, clients may have diverse distributions of images, including variations in signal-to-noise ratio, brightness, and disease prevalence, causing representation gaps between each client and divergence when aggregating [8]. Furthermore, the requirement for constant communication with the central server in FL can pose challenging for certain medical institutions. Although one-shot or few-shot FL is under research [9], this approach has not been fully validated.

Previous studies mainly focused on comparing FL models with each other and used popular benchmark datasets such as CIFAR10 or BraTS [10,11]. This study compared the efficacy of two cross-silo image segmentation FL models with industry-level CL models under similar conditions, employing a large-sample sized pragmatic non-i.i.d. brain MRI dataset while showing that FL does not significantly degrade performance compared to CL. We compared FL models employing constant communication with those employing one-shot aggregation. To the best of our knowledge, our scale of FL experiments have not been investigated extensively, especially in the biomedical applications. Additionally, we introduce a “batch clipping” technique, which limits the number of samples per client each epoch and accelerates the FL training process with negligible performance trade-off.

2. Materials and Methods

Two datasets are used in this retrospective research: the acute ischemic lesion dataset (Task I) and the chronic ischemic lesion dataset (Task II). The datasets were used in previous works concerning the developments of the two CL models introduced in this paper. To compare the performance of a commercially used CL-based algorithm with an FL-trained algorithm, we adopted the CL architecture of the commercial software and a corresponding FL architecture for this comparison. All data shown below are utilized identically in both FL and CL experiments. Written informed consent was provided by subjects or their legal representatives. All examinations were done in accordance with the 1964 declaration of Helsinki, and the local ethics committee (institutional review board of Dongguk University Ilsan Hospital).

2.1. Dataset and brain image analysis

Acute ischemic lesion dataset From May 2011 to November 2013,

we consecutively enrolled 14,740 subjects with ischemic stroke or transient ischemic attack admitted to 10 participating centers within seven days from symptom onset (Fig. 1). After excluding 1194 subjects using the following criteria: contraindication to MRI ($n = 688$), poor quality or unavailability of brain diffusion-weighted imaging (DWI) ($n = 321$), and MRI registration error ($n = 185$), leaving 13,546 subjects amongst 9 institutions for training and validation and 1 institution for external testing. Previous study [12] for CL utilized a total of 13,597 subjects originated from 10 institutions. Brain DWIs were obtained using 1.5-T ($n = 9084$) or 3.0-T ($n = 2923$) MRI systems. DWI protocols were b-values of 0 and 1000 s/mm^2 , TR of 2400–9000 ms, TE of 50–99 ms, voxel size of $1 \times 1 \times 3\text{--}5 \text{ mm}^3$, interslice gap of 0–2 mm, and thickness of 3–7 mm. Ischemic lesions on DWIs were segmented by six experienced researchers using an in-house software under the close guidance of an experienced vascular neurologist as previously reported [13–15].

Chronic ischemic lesion dataset From May 2011 to November 2013, we consecutively enrolled 10,423 subjects admitted to 9 participating centers within seven days from symptom onset. We excluded the following subjects: contraindication to MRI ($n = 315$), poor quality or unavailability of brain diffusion-weighted or fluid-attenuated inversion recovery (FLAIR) MRI ($n = 1632$), and MRI registration error ($n = 55$), leaving 8421 subjects amongst 3 institutions for training and 6 institutions for testing. All data were forwarded from the previous CL study [16]. Brain FLAIR MRI was performed on 1.5 T ($n = 6583$) or 3.0 T ($n = 1803$) MRI systems. FLAIR image protocols were TE 76–187 ms, TR 6000–11568 ms, voxel size $1 \times 1 \times 3\text{--}7 \text{ mm}^3$, spacing 0.3–1.0 mm, slice thickness 3–7 mm, FOV 175–280 mm, and matrix size (row) 256–768. As previously reported [15,17,18], each subject’s high signal intensity white matter lesions on FLAIR were manually segmented by six experienced researchers under careful supervision by an experienced vascular neurologist. When chronic white matter lesions on FLAIR and acute infarct lesions on DWI overlapped or were adjacent, we determined the extent and distribution of FLAIR WMH based on lesions in the hemisphere contralateral to the acute infarct location, as WMH symmetry of morphology and distribution are often observed between hemispheres [15,17,18].

2.2. Image preprocessing

Acute ischemic lesion dataset Brain DWIs were preprocessed by applying Otsu’s threshold and N4 bias correction [19,20]. We adjusted DWI signals to fit between 0 and 255 before applying z-score normalization. Each image has been augmented with a linear combination of slice-wise similarity transform, MRI bias field simulation, axis flip, and gamma/contrast change.

Chronic ischemic lesion dataset We applied 2D B-spline interpolation to resize the FLAIR slices to a dimension of 256×256 pixels. Next, we performed slice-wise intensity normalization in a uint8 format to ensure pixel intensities ranged from 0 to 255 on each slice. We subsequently performed case-wise intensity equalization to ensure comparability in the imaging data across different MRI vendors by employing a histogram of 32 bins and shifting the highest peak of the histogram value to 150. Then, we generated a binary brain mask by including all voxels with signal intensities greater than a threshold value of 30 in the histogram domain. We filled tiny holes in this brain mask using binary fill holes from `scipy.ndimage`. Finally, we performed the Gaussian normalization process to yield pixel intensities with a mean of 0 and a standard deviation of 1 for the brain area under the brain mask, producing optimal arrays for training a deep learning model.

2.3. Batch clipping

The largest client possessed three times more data than the smallest, leading to inefficiencies in training, where small clients were kept idle for extended periods, awaiting one or two large clients. To reduce the total training time, we adopted a “batch clipping” strategy that caps the

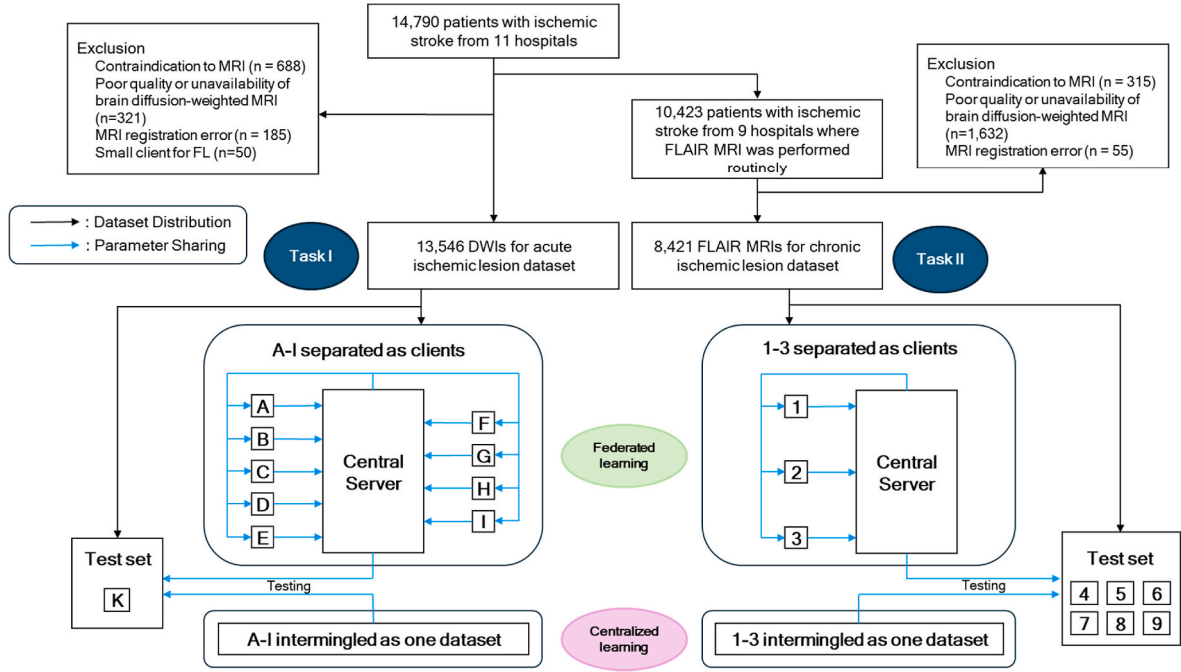


Fig. 1. Study flow chart and study scheme for Task I, II. Blue arrows indicate the flow of model parameters, disseminated from the central server, trained by each client, and submitted to the server again. Exact distribution of the data can be found in the supplementary materials.

maximum number of batches per epoch. Even if certain data may not appear in one epoch, the expectation is that through the random batch shuffling over sufficient iterations, most data points will eventually be included in the training process. We constrained the number of batches to 200 and 1250 for Task I and II respectively (Fig. 2), which corresponds to approximately 50~55 % of the largest client's batch count. Clients with fewer batches than the clipping threshold remain unaffected, utilizing their entire dataset in every epoch. We experimented both tasks with and without batch clipping to investigate its effect.

2.4. Implementation details

FL models in both tasks are compared with an existing CL model with industry-level performance. The choice of ML model, number of epochs, hyperparameters, optimizers, and loss functions are matched as closely as possible with previous studies (Details in the Supplemental Digital Content Document). To minimize communication overhead between server and clients, we implemented four epochs per round, indicating aggregation of results after 4 epoch of training in each client. After aggregation, each client validates the aggregated model on their own validation dataset. We conducted both experiments using three RTX A6000 GPUs, with Flower [21] as the common FL framework.

We employed two aggregation methods in our study: FedAvg [5] and FedYogi [22]. FedAvg is a straightforward method that averages the values of all received models on a coordinate-by-coordinate basis. FedYogi employs the Yogi [23] optimizer on the server side for more efficient updates and represents a more advanced variant of FedAvg [24]. The two methods were chosen based on their flexibility as baseline procedures for other more sophisticated methods. We expected an FL model performing well with minimum complexity being comparable with CL would lend a stronger justification for the feasibility for FL in general. Moreover, we did not select any method that significantly alters the client-side update rule from traditional CL training, as it could potentially interfere with our implementation of batch clipping.

Task I segmented the acute ischemic lesion on diffusion-weighted image (DWI) using STU-Net [25]. We partitioned the dataset into training and validation sets with a random sampling of a 4:1 ratio. The external test dataset for Task I consisted of 2777 DWIs acquired from a single institution not included in the training dataset. As the CL model for Task I utilized 1000 epochs for full convergence, we elected FedYogi for the aggregation method, which was expected to perform better than FedAvg over an elongated training. Both FL and CL models use PyTorch [26] as the ML framework. When evaluating the test Dice similarity coefficient (DSCs) over multiple rounds, we selected the first 100 DWIs

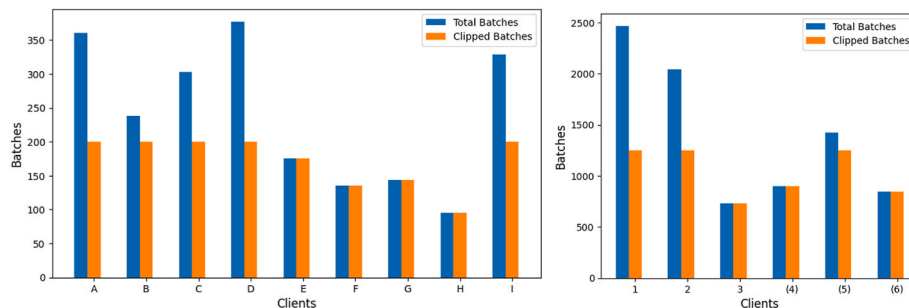


Fig. 2. The Number of Total and Clipped Batches for Task I and II. The blue bar shows the number of batches of the entire dataset per client, and the red bar shows the clipped number of batches. Clipped clients stop training each epoch once it trains up to the clipping value. Clipping values are 200 for Task I and 1250 for Task II. Clients possessing less batches than the clipping value are unaffected.

of the external test dataset due to time constraints. Final DSC results were calculated with the entire external test dataset.

Task II segmented the chronic ischemic lesion, known as white matter hyperintensities in FLAIR MRI, using U-Net [27]. Task II dataset used the identical partition as the CL model, a 3:1:1 ratio for training, validation, and internal testing. However, we did not use the internal testing dataset for our experiments but rather used the external test datasets from another six institutions, ranging from 8422 to 53344 slices per institution. In the Task II, we elected FedAvg as the aggregation method to show FL convergence without extensive hyperparameter tuning. Both FL and CL models use Tensorflow [28] as the ML framework. When evaluating the test DSCs over epochs 2–12, we selected only the first 2000 slices of the test set due to time constraints. The final DSC results were calculated with the entire test set.

Additional studies on Task II repositioned three institutions from the external test dataset to clients for training to prove the robustness of FL with varying clients. The three parenthesized clients in Fig. 1 are the three clients excluded from the primary FL test results but involved in the additional study. All settings are held identical to the original Task II experiments.

2.5. Statistical analysis

To compare demographic and imaging characteristics between training and external test datasets, we used *t*-test, rank-sum test, ANOVA, and Kruskal Wallis for continuous variables and chi-square test for categorical variables as appropriate.

To test the model performance in Task I and II, we used the DSC. All DSCs shown in the tables are averages and standard deviations of the dataset specified. For analyzing the main results, we employed paired sample *t*-tests per client to compare FL and CL models. In Task II, we utilized one-way ANOVA tests to compare DSC in each client in the external test dataset. For both statistical analyses, the *p*-value for significance is defined as less than 0.05. All calculations were performed using the `scipy.stats(= 1.10.1)` library in Python(version 3.8.10, The Python Software Foundation).

3. Results

3.1. Patient demographics

For Task I, the mean (SD) ages for the training & validation, and external test datasets were 68.1 (12.8) and 68.2 (12.4), respectively. There were significant differences in demographic and imaging characteristics between the training and external test datasets (Table S1 in the Supplemental Digital Content Document). For Task II, the mean (SD) age for the training & validation was 67.4 (13.0) and 60.4 % were men (Table S2 in the Supplemental Digital Content Document). Mean ages for external test datasets ranged from 67.5 to 70.0. All variables

significantly differed across the datasets, indicating dataset heterogeneity.

3.2. Main test results

For Task I, the FedYogi FL models, with or without batch clipping, trained for 90 rounds (equivalent to 360 epochs) achieved a DSC of 0.754 ± 0.183 and 0.762 ± 0.173 , respectively (Table 1). These results are better than those of the CL model trained for 360 epochs (DSC 0.691 ± 0.229 ; $p < 0.001$) and are comparable to the CL model trained for 940 epochs (DSC 0.755 ± 0.207 ; $p = 0.701$), which is the best-performing CL model out of 1000 epochs. In turn, the 250-round (1000 epochs) trained FL models, with or without batch clipping exhibited a DSC of 0.776 ± 0.170 and 0.776 ± 0.169 , respectively, outperforming the 940-epoch CL model ($p < 0.001$). For Task II, segmenting chronic ischemic lesions, the FL Clipping, FL No Clipping, and CL models all performed similarly after 48 epochs (0.751 ± 0.304 , 0.761 ± 0.299 , 0.744 ± 0.304).

For comparison, we also conducted few-shot FL experiments that aggregate only once or twice at the end of extensive decentralized learning. For Task I, a single round of FedYogi aggregation after training 132 epochs was not trained properly (DSC 0.142 ± 0.184 ; $p < 0.001$ compared with FL model with 37 rounds (148 epochs)). Task II tested a two-round, 24 epochs per round FedAvg FL, for an equivalent total epoch of 48. The result is significantly lower than FL with our standard four epochs per round (DSC 0.581 ± 0.437 ; $p < 0.001$).

3.3. Task I convergence

Fig. 3a and b depict training and validation losses for Task I. The CL model and all clipped FL clients properly converge while training, albeit at different values. The training and validation losses of FL without batch clipping is included in Fig. S1 in the Supplemental Digital Content Document. By the end of 1000 epochs, the CL model reaches 0.121 training loss (Fig. 3a), while FL clients range from 0.113 to 0.182 in each training client with a weighted average of 0.144. For the validation loss (Fig. 3b), the CL model reaches 0.126, whereas FL clients range from 0.098 to 0.176 with a weighted average of 0.138. Fig. 3c compares the Test DSC for FL and CL models throughout their training and includes a smoothed-out graph for convenience. While the average FL losses were consistently higher than CL losses, FL consistently outperforms CL after 200 epochs in the test dataset, converging to a higher DSC by the end of the training.

3.4. Task II convergence

Fig. 4a and b shows the validation DSC for each FL training client in Task II, with and without batch clipping. The test DSC scores are lower than the validation DSC as the test set originates from new clients and possibly has a different distribution. Fig. 4c compares the test DSC of the

Table 1
Comparison of segmenting performance between federated and centralized learnings.

Model	Acute ischemic lesion								Chronic ischemic lesion			
	STU-Net								U-Net			
Number of clients for training	9								3			
Aggregation method	FL, FedYogi								FL, FedAvg			
Batch Clipping	Yes								Yes			
	No								No			
	N/A								N/A			
Rounds ^a	33	90	250	1	90	250	N/A	N/A	12	2	12	N/A
Epochs	132	360	1000	132	360	1000	360	940	48	48	48	48
Avg. DSC ^b	0.717	0.754	0.776	0.142	0.762	0.776	0.691	0.755	0.751	0.581	0.761	0.744
SD ^b	0.199	0.183	0.170	0.184	0.173	0.169	0.229	0.207	0.304	0.437	0.299	0.304

FL = federated learning; CL = centralized learning; N/A = not available; DSC = Dice similarity coefficient; SD = standard deviation.

^a After each round, the results of training were sent to the server, aggregated by FedYogi or FedAvg methods and returned to each client.

^b Average DSC and standard deviation were calculated using DSC per patient.

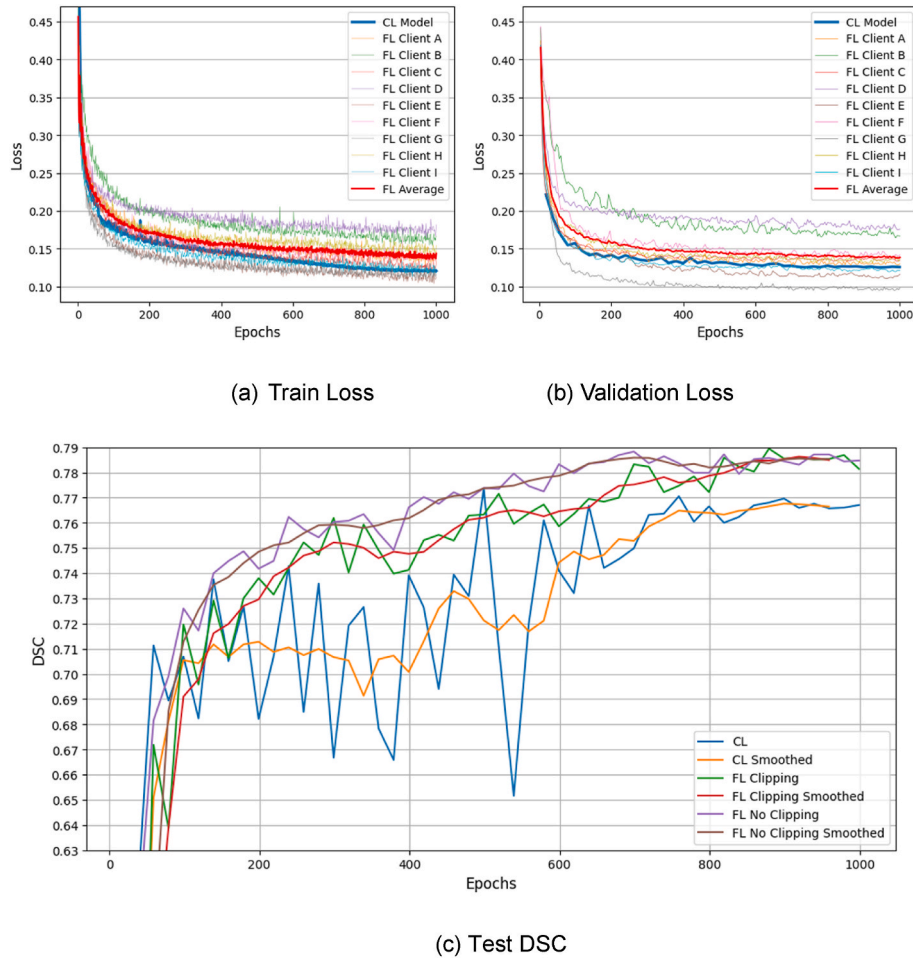


Fig. 3. STU-Net Task I Losses and Test DSCs. a) Training loss of the CL model and clipped FL clients 0–8, up to 1000 epochs. b) Validation Loss of the CL model and clipped FL clients 0–8, up to 1000 epochs. Validation was performed every 4 epochs after aggregation. c) DSC of the CL model and aggregated FL models evaluated with the first 100 test set images, up to 1000 epochs. A smoothed out version using convolution with 5 previous results are drawn for ease of comparison.

Clipping, No Clipping, and CL models. All three models converge to ~ 0.75 DSC by the end of 48 epochs (equivalent to 12 rounds). Table 2 summarizes the average test DSC per each client in the external test dataset. While there are disparities amongst test clients, a one-way ANOVA test with the average DSC results does not reveal any statistical significance in their variances ($p = 0.650$). The FL model without clipping took around 3.5 h to train 48 epochs, whereas FL with clipping took only about 1.75 h.

3.5. Additional studies: task II training with six clients

To directly assess the impact of training variability and confirm that our results were not dependent on an arbitrary selection of clients, we conducted an additional study for Task II. In this experiment, we altered the composition of the training and testing cohorts by incorporating three institutions (Clients 4, 5, and 6) from the original external test set into the training process (Fig. S2 in the Supplemental Digital Content Document). While individual performance shifted among the remaining test clients (Table S3 in the Supplemental Digital Content Document), the overall outcome remained consistent. A one-way ANOVA test of the average Dice Similarity Coefficient (DSC) across the external test centers revealed that the performance differences were not statistically significant ($p = 0.846$). This demonstrates the FL model's robustness to significant variations in the training set and indicates that its effectiveness is not tied to one specific client configuration.

The second additional study experiments batch clipping in CL, concatenating the datasets used in FL batch clipped training to form one

centralized dataset, on both three and six client settings (Fig. S5 in the Supplemental Digital Content Document). Test DSCs were calculated identically to the original Task II experiments. Results show that CL with batch clipping does not perform as well as FL with batch clipping, showing slower convergence speed and lower final DSC.

The third additional study shows test DSC for models trained individually by each client (Fig. S6 in the Supplemental Digital Content Document). Client 5, the best performing client, shows performance comparable to FL models (0.749), but other five clients fail to reach 0.73 DSC by the end of 48 epochs, implying that most clients would be motivated to participate in FL training instead of training their own model. The number of data for each client heavily effects the effectiveness of individual training, except for client 5.

4. Discussion

This study compared the performance of FL and CL models, demonstrating the practicality of FL in segmenting acute and chronic ischemic brain lesions. We experimented in both data-intensive (Task A) and a relatively lightweight (Task B) tasks to show the generalizability of FL in both spectrums. In addition, we demonstrated that one-shot or few-shot communication approach yielded lower performance of FL models compared with constant communication. We also empirically demonstrated that batch clipping is a valid strategy for reducing training time without sacrificing robustness.

The BraTS dataset—a benchmark for brain image segmentation distributed across multiple centers—has been utilized in medical FL

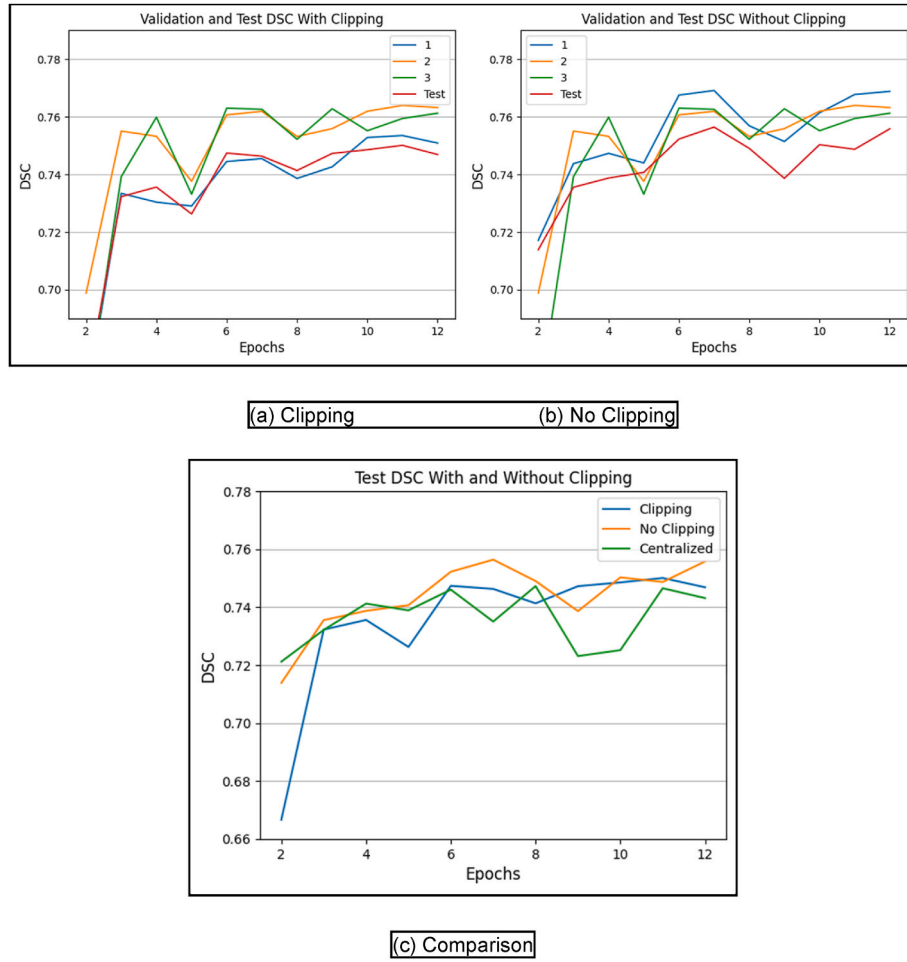


Fig. 4. Task II Validation and Test DSCs. a) Validation DSC for clients 1–3 and Test DSC of the aggregated central model for rounds 2–12 with batch clipping. b) Validation DSC for clients 1–3 and Test DSC of the aggregated central model for rounds 2–12 without batch clipping. c) Comparison of the Test DSC for Clipping, No Clipping, and Centralized models. All Test DSCs in this figure were calculated using the first 2000 slices of the test set. All graphs of Task II start from the second round since FedAvg produces values close to 0.1 for validation and tests in the first aggregation.

Table 2

Task II average Dice similarity coefficient for each client in external test datasets.

Client ^a	Clipping	No Clipping	Centralized
	12 Round	12 Round	48 Epoch
4	0.735	0.737	0.724
5	0.704	0.711	0.704
6	0.743	0.757	0.739
7	0.719	0.727	0.710
8	0.763	0.774	0.755
9	0.779	0.790	0.772
Total	0.751	0.761	0.744
SD ^b	0.0254	0.0271	0.0243

SD = standard deviation.

^a Each client represents a separate institution.

^b Standard deviation was calculated using DSC per center.

studies due to its partition by medical centers. Pati et al. [3] trained on BraTS and other datasets to produce a state-of-the-art FL model. However, they did not start their training from a randomly initialized model but rather a “public initial model” pre-trained from the BraTS dataset. Li et al. [10] also used BraTS to simulate the non-i.i.d nature of medical data and empirically shows multiple experiments for differential privacy. The scale (~10,000 images) of our datasets proves that FL functions in larger datasets than BraTS, even with a randomly initialized model.

Our findings suggest that FL necessitates constant communication between the clients and the central server, rather than relying solely on one or two rounds of aggregation after multiple epochs of individual training. Although a two-round aggregation for Task II yields reasonable performance, it lacks generalizability beyond each client’s training set, resulting in unsatisfactory final performance compared to FL models that communicate every four epochs. FedYogi and FedAvg struggle to effectively aggregate all parameters coherently in a single round, leading to poor performance after just a few rounds of aggregation.

Implementing batch clipping effectively reduced the training time by half compared to models without it. In FL, where training occurs in parallel across multiple institutions, the overall duration is dictated by the slowest client in each round rather than the total data size. Therefore, limiting the maximum number of batches trained for each epoch can reduce the total training time. Moreover, despite an increase in the number of clients in Task II for additional study, the training time remained relatively constant for FL, showcasing its parallelized nature. In contrast, the training time of CL increased linearly with the size of the training data, consuming round 6.5 h. However, CL does not benefit from batch clipping as well as FL. While the training time significantly decreases, the reduction in size of the aggregated training data affects the training quality much more than FL.

We attribute the efficacy of batch clipping to the aggregation methods employed in our experiments—FedYogi and FedAvg. Both methods calculate the weighted average of updates, considering the

number of batches processed by each client in every round. Thus, even though some clients train on less data per epoch, the norm of the total updated gradient remains balanced as updates from other clients are more heavily weighted. Although not included in this investigation, we anticipate that batch clipping should be compatible with other aggregation algorithms such as KRUM [29] and median, as well as defensive mechanisms like Differential Privacy [30].

However, it should be noted that our implementation of batch clipping is not a standard procedure in theory. In most widespread aggregation methods, it is normal to take the average weighted values of each client's submitted values based on their dataset size, not their number of updates. However, batch clipping pretends as if, in the current round, that there are only at maximum a certain number of data points for any client, which are also sampled randomly. Our batch clipping scheme would therefore lead to a heterogeneous number of updates for each sample, depending on whether it is a part of a larger dataset or a smaller one. This is where empirical results deviate slightly from theory, as it is difficult to neither prove nor disprove that such change would actively harm the FL training in the long term. In this specific set of experiments, there were no signs of degradation caused by batch clipping; nevertheless, we recognize that batch clipping might be subject to further scrutiny in future FL experimental designs. A simple yet effective solution for the imbalance in the number of updates between samples could be to adjust the probability of client selection rather than a uniform distribution such that each sample will, in expectation, be updated an equal number of times.

In Task I, the FL model performed better than the CL model in testing scenarios, which is not typical. In most studies, FL is considered to have a strict performance ceiling below CL [31]. The expectation holds true in training and validation losses, where even though some clients score lower training and validation losses, the average FL loss is consistently greater than the CL loss. However, the potential reason behind the reversal of performance in the test set needs to be addressed. First, it could be that the averaging nature of FL could cancel out erratic updates in opposite directions. Indeed, Karimireddy et al. [32] suggests “bucketing,” which involves averaging some client submissions before aggregating to reduce variance between clients. The other possibility is that the data of smaller clients in the training dataset happened to represent the test data better than those the larger clients, and hence act as a “well-represented” dataset. In CL, the updates induced by small amount of well-represented data could be diluted by data from different institutions. Even though FL still takes a weighted average of the number of data for each client when aggregating, the small client can train solely with its well-represented data, which can steer the aggregated model towards higher test scores regardless of training and validation losses. Indeed, there were some correlations between the small, performant clients such as client E and G and the external test dataset such as lesion area and slice thickness, depicted in Figs. S3 and S4 in the Supplemental Digital Content Document.

In the third additional study, client 5 performed better than the FL model including it. We did not, however, consider client 5's data to have “more generalization” on the test data. Client 5 showed significantly less DSC compared to other clients for the FL model only trained on clients 1–3, which suggests that it has a distribution far from all other clients. A feasible explanation from our understanding is that client 5 might have a more heterogeneous distribution of data than other clients. This could have induced a low score when it was a part of the test data set but rather acted as a positive quality when trained separately.

We acknowledge several limitations of the study. Most notably, all experiments were conducted in simulated environments, as our motivation was to translate an existing CL model into an FL model. Therefore, all data were preprocessed identically and maintained as high-quality, pixel-level segmentation data. In practice, such ideal conditions may need to be relaxed [33]. Also, some minor assumptions were made during the design of FL models. For instance, the batch size is set to 4 instead of 12 for the original Task I CL model due to GPU memory

constraints during simulating. The choice to set four epochs per round was arbitrary and can be adjusted to any other reasonably small value. We believe these assumptions have minimal impact on the major findings of our experiments; however, they are reported for potential investigations. Finally, the trials in Task I were not repeated multiple times nor with varying training sets due to constraints in time and resources. While we did find statistical significance in FL performance being at least comparable to CL in Task I, we encourage future experiments to extend our results for additional confirmation of our findings on large-scale FL.

In conclusion, our study provides two FL brain lesion segmentation models that rival CL models, supporting the suitability of FL in medical image segmentation. Throughout our experimentations, we were able to utilize batch clipping for efficient training. Our findings may facilitate future studies involving a larger set of clients and more diverse datasets.

CRediT authorship contribution statement

Joon Kim: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Hoyeon Lee:** Writing – review & editing, Supervision, Software, Methodology, Data curation. **Jonghyeok Park:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Data curation. **Sang Hyun Park:** Writing – review & editing. **Myungjae Lee:** Writing – review & editing. **Leonard Sunwoo:** Writing – review & editing. **Chi Kyung Kim:** Writing – review & editing. **Beom Joon Kim:** Writing – review & editing. **Dong-Eog Kim:** Writing – review & editing. **Wi-Sun Ryu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Data sharing statement

Data generated or analyzed during the study are available from the corresponding author by request.

Ethics statement

Written informed consent was provided by patients or their legal representatives. All examinations were done in accordance with the 1964 declaration of Helsinki, and the local ethics committee (institutional review board of Dongguk University Ilsan Hospital).

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hoyeon Lee, Jonghyeok Park, Myungjae Lee, and Wi-Sun Ryu are employees of JLK Inc., Seoul, Republic of Korea.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ibmed.2025.100283>.

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